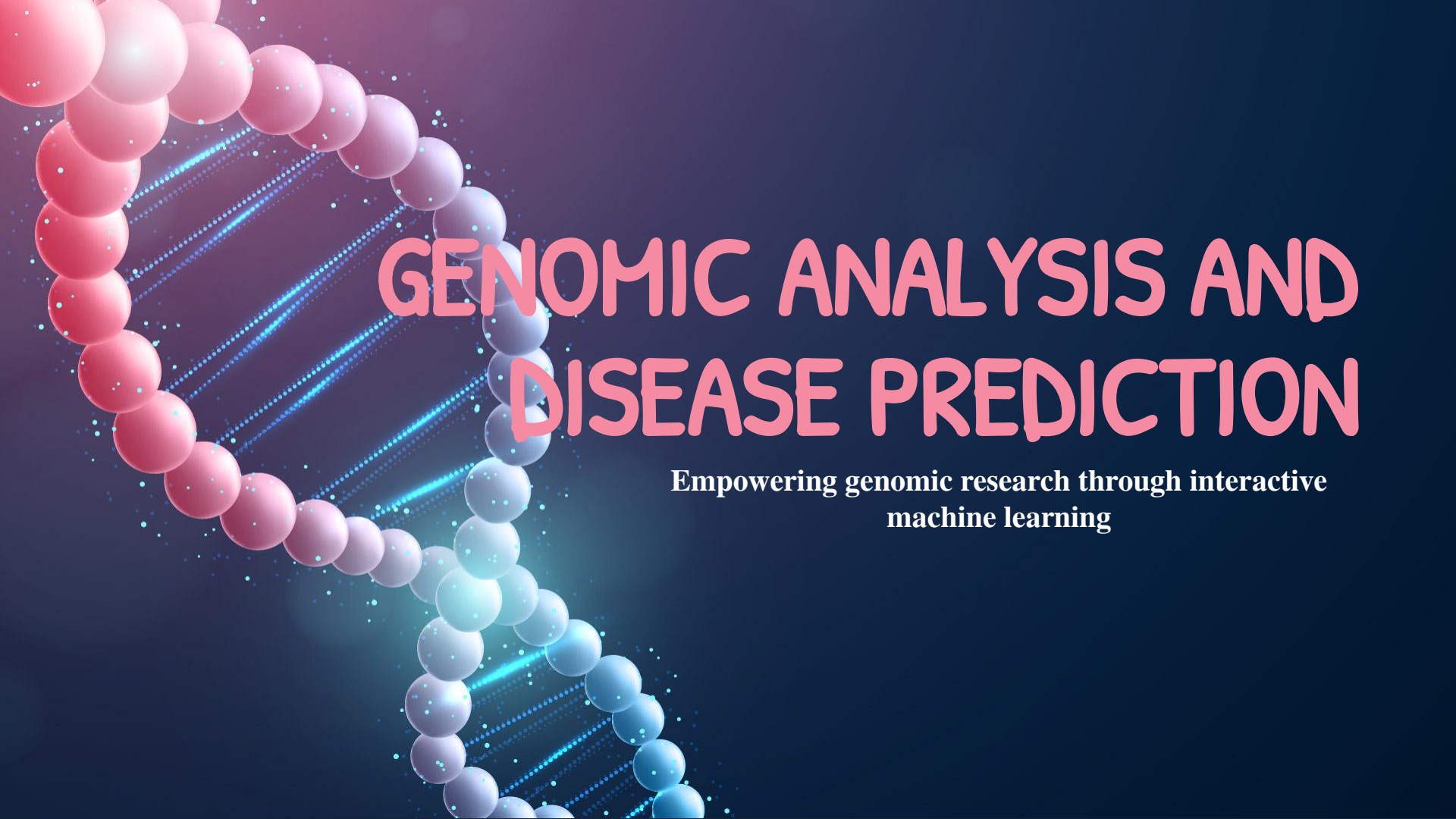




GENOMIC ANALYSIS AND DISEASE PREDICTION

Empowering genomic research through interactive
machine learning

INTRODUCTION

- Genomic data science is a field of study that enables the researchers to use powerful and computational and statistical methods to decode the functional information hidden in the DNA sequence.
 - Estimates predict that the genomic research will generate between 2 and 40 exabytes of data within the next decade.
 - Our ability to sequence DNA has far outpaced our ability to decipher the information it contains so genomic data science will be a vibrant field of research for many years to come.
 - Performing genomic data science carries with it a set of ethical responsibilities as each person's sequence data are associated with issues related to privacy and identity.
- 



1

PURPOSE

An interactive tool for analyzing genomic data using machine learning

2

TARGET USERS

Researchers,educators,student in genomics and bioinformatics.

3

TECHNOLOGY

Python-based,runs in google collab environment.

4

CAPABILITIES

Data Analysis , model training , disease prediction and visualization

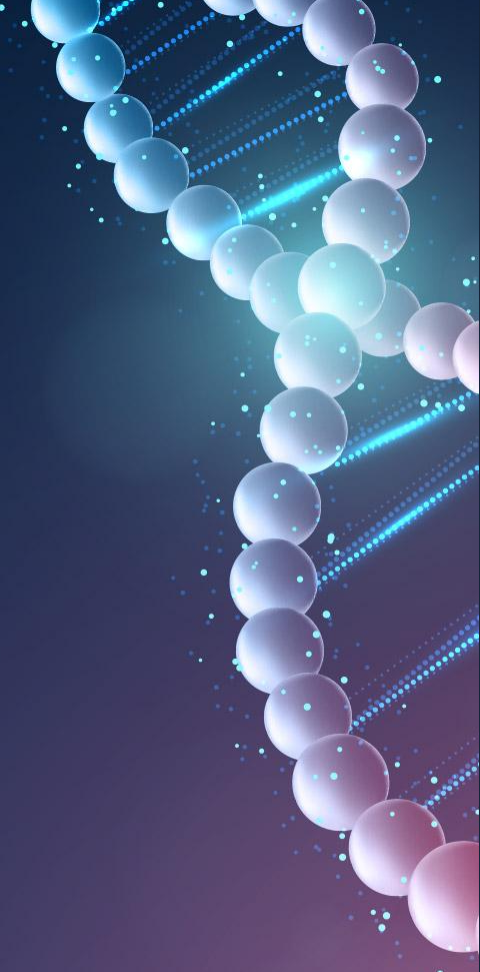


KEY FEATURES

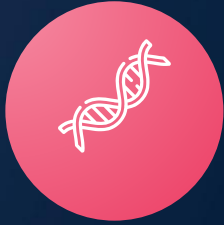
- Upload custom genomic data or use sample datasets
- Train multiple machine learning models simultaneously
- Compare model performance with detailed metrics
- Identify important genomic markers
- Predict disease risk based on genomic profiles
- Generate visual reports and interpretations

DATA REQUIREMENTS

- Format: CSV files with genomic markers (SNPs) as features
- Content
 - Columns: Multiple SNP values (typically encoded as 0,1, 2)
 - Last column: Target phenotype or disease classification
- - Classification Types :
 - Binary: 0 (healthy) vs 1 (disease)
 - Multi-class: Multiple disease types (0, 1, 2, 3, etc.)



MACHINE LEARNING MODELS



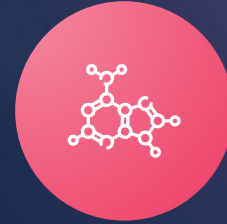
RANDOM FOREST

- Ensemble method using multiple decision trees
- Provides feature importance ranking



SUPPORT VECTOR MACHINE

- Effective for linearly and nonlinearly separable data
- Multiple kernel options: RBF, linear, polynomial, sigmoid

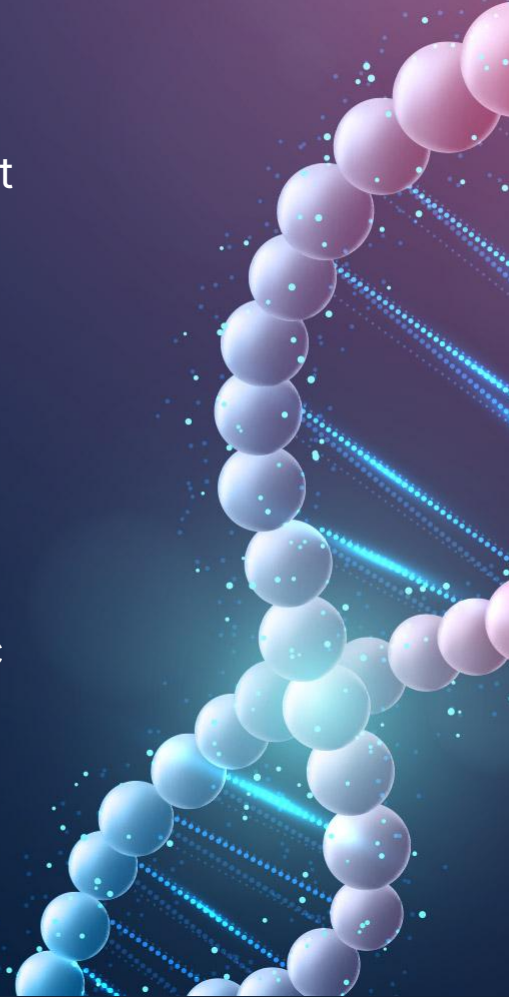


NEURAL NETWORK

- Multi-layer perceptron architecture
- Customizable hidden layer sizes

WORK FLOW

- Data Upload: User uploads genomic data in CSV format
- Configuration: User selects models and parameters
- Training: Models are trained on a portion of the data
- Evaluation: Performance metrics are calculated and visualized
- Disease Prediction: Risk assessment based on genomic markers
- Results: Interactive visualizations and summaries are presented



DATA ANALYSIS COMPONENTS



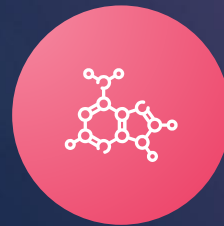
DATA INFORMATION

- Overview of samples, features, and class distribution



MODEL PERFORMANCE

- Accuracy, precision, recall, F1 score for each model



VISUALIZATIONS

- Model performance comparison charts
- Confusion matrices
- Feature importance
- ROC curves (for binary classification)

DISEASE PREDICTION PROCESS

Visual Risk Assessment

Horizontal bar charts showing risk percentages

Risk Summary Table and level classification

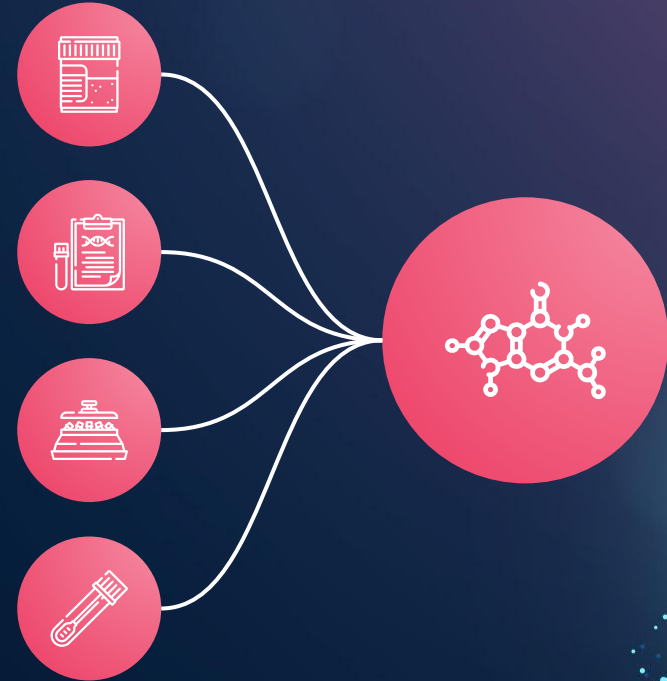
Detailed probability for each disease class
High, Moderate, Low, or Very Low

Recommendations

General advice based on risk level

Accuracy Validation

Comparison with actual disease class



ADVANTAGES OF THE TOOL



Ease of Use

No coding required for basic analysis



Reproducibility

Consistent results with same random seed



Flexibility

Multiple models and parameters can be adjusted



Extensibility

Foundation for more advanced analyses



Interpretability

Clear visualizations and explanations



Scalability

Will take less time when implemented on cloud

TECHNICAL IMPLEMENTATION



scikit-learn

Machine learning
algorithms

pandas

Data manipulation

**matplotlib
/seaborn**

Visualization

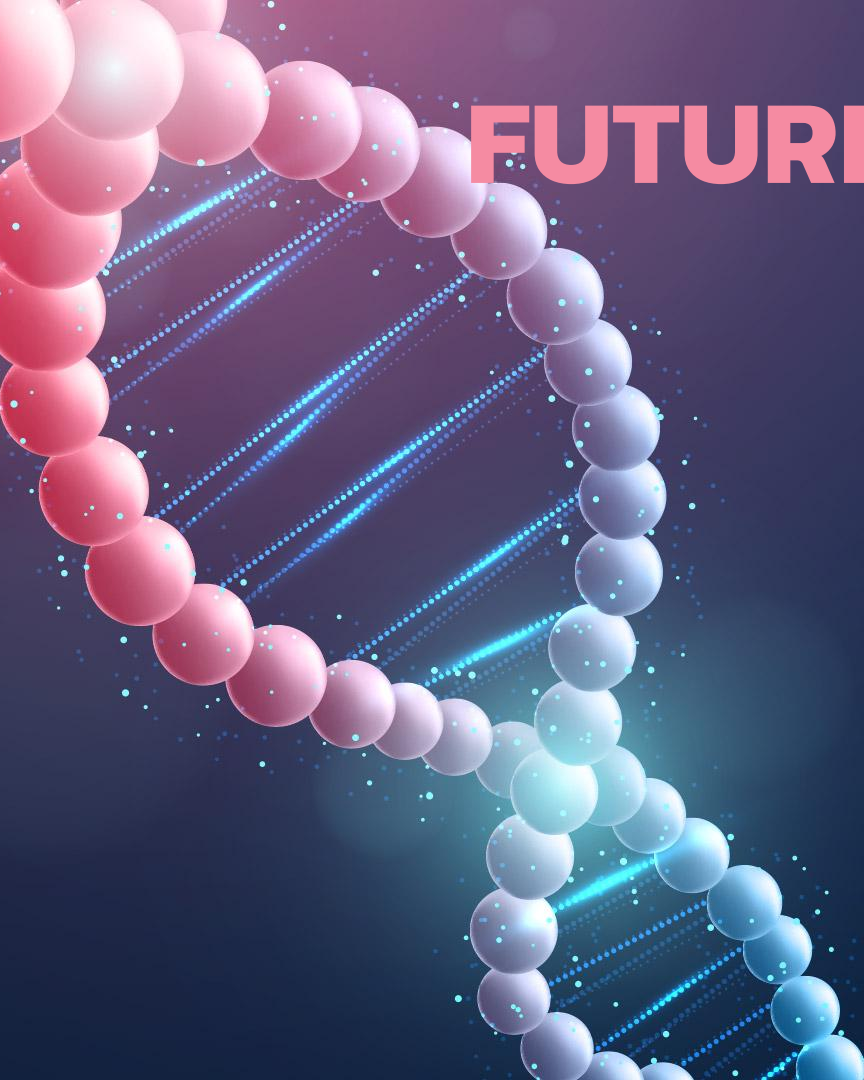
ipywidgets

Interactive UI
components

LIMITATIONS AND CONSIDERATION

- Educational Purpose: Not for clinical decision-making
- Data Quality: Results depend on quality and size of input data
- Validation Needed: Real applications require rigorous validation
- Computational Constraints: Limited by Colab environment resources
- Ethical Considerations: Genomic data requires appropriate handling





FUTURE ENHANCEMENTS

- New Sample Prediction: Direct input of new genomic profiles
- Feature Selection: Automated identification of optimal markers
- Advanced Models: Integration of more sophisticated algorithms
- Cross-Validation: More robust performance evaluation
- Integration: Connection with existing genomic databases
- Implementation on AWS: More reliable and scalable with faster processing of the pipeline with a large datasets.

CONCLUSION



- Powerful tool for exploratory analysis of genomic data
- Demonstrates the potential of machine learning in genomic medicine
- Provides educational value for understanding genotype-phenotype relationships
- Foundation for more advanced research and applications
- Bridges the gap between genomic data and actionable insights



THANK YOU

